Section 18 Report Drafting Guide
Annual Climate and Year-Round Operating Layer

April 19, 2026

1 Purpose

This document is a report-writing guide for Section 18 of `docs/execution_docs/physics/physics_and_math_ref` ("Annual climate and year-round operating layer", Eqs. 229–251). It maps each variable from the equations to:

- its JSON source in `Heat Transfer Study/study_config.json`,
- its runtime variable in `Heat Transfer Study/heat_transfer_study.py`,
- and the exported outputs in `Heat Transfer Study/outputs/year_round/study_summary.txt`.

2 Identity Clarifications For Implicit Symbols

To remove ambiguity in the Section 18 equations:

- m: month index (1 to 12).
- d: day-of-year evaluation index (1 to P).
- P: annual day period ($P = 365$ in the present implementation).
- d_i: day-of-year reference day for reference point i (month midpoint in current implementation).
- d_m: month-midpoint day-of-year for month m. With one reference point per month, $d_m \equiv d_i$.
- L_m: number of days in month m from `daysPerMonth`.
- i in Eqs. (229)–(230): index over monthly reference points.
- i in Eqs. (232), (236): day-of-month candidate index inside one month for event placement. This is a different role from kernel index i.
- y_i: selected reference-value family at reference point i. It is one of:
  - $y_i = \bar{T}_i$ for the daily baseline ambient regression pass,
  - $y_i = \sigma_i$ for the daily ambient-spread regression pass.
- $\hat{y}(d)$: kernel-regressed daily estimate from the active y_i pass.
- $w_i(d)$: Gaussian kernel weight for reference point i at day d.

- h: Gaussian kernel bandwidth in days.
- $\mathrm{LR}(\cdot)$: largest-remainder operator used to convert expected monthly event counts to integer allocations.
- $s_{f,m}$, $r_{f,m}$, $\mathcal{S}_{f,m}$: freeze placement spread, repulsion length scale, and selected-day set for month m.
- $s_{h,m}$, $r_{h,m}$, $\mathcal{S}_{h,m}$: heat-wave start placement spread, repulsion length scale, and selected-start set for month m.
- j: heat-wave spell index within month m, ordered by start day.
- $\Delta T_{h,m}^{\mathrm{low}}, \Delta T_{h,m}^{\mathrm{high}}$: NOAA-derived lower/upper bounds for heat-wave spell anomaly above $\bar{T}_m$.
- $\sigma_{\Delta T,h,m}$: configured or fallback spread for bounded spell-anomaly sampling in month m.
- $\Delta T_{h,m,j}$: sampled anomaly assigned to spell j, bounded to $[\Delta T_{h,m}^{\mathrm{low}}, \Delta T_{h,m}^{\mathrm{high}}]$.
- f_{day}: generic clipped-duty function in Eq. (244).
- $\dot{Q}_{\mathrm{load}}$, $\dot{Q}_{\mathrm{nom}}$: generic load/capacity placeholders inside Eq. (244); these map to heating or cooling quantities depending on active mode.
- $D(d)$ in Eq. (251): active duty fraction for the day, equal to $D_h(d)$ on heating days or $D_c(d)$ on cooling days.

3 Temperature Variables: Explicit Definitions and Sources

- $\bar{T}_m$: monthly Connecticut mean ambient from `climate_data.monthlyMean_C`.
- σ_m: monthly ambient spread from `climate_data.monthlyStd_C`.
- $T_{\infty,\mathrm{base}}(d)$: periodic Gaussian-kernel regression of $\bar{T}_m$ over day-of-year.
- $\sigma(d)$: periodic Gaussian-kernel regression of σ_m over day-of-year.
- $T_{\infty,\mathrm{freeze}}(d)$: intermediate profile after freeze-day overrides.
- $T_m(d)$: month-local ambient profile during event-overlay construction.
- $T_\infty(d)$: final day ambient used in load/capacity re-solves.
- $T_{h,\mathrm{set}}, T_{c,\mathrm{set}}$: resolved heating/cooling setpoints from annual config (or inherited design defaults).
- $T_b(d), T_t(d)$: bottom/top bed-node temperatures from Eq. (247), solved using $Q_{\mathrm{bed}}(d)$ and $T_\infty(d)$.
- Heat-wave spell threshold temperature: $\bar{T}_m + \Delta T_{h,m,j}$ for spell j in month m.
- NOAA derivation artifacts for spell-anomaly bounds:
  - `Heat Transfer Study/data/climate/noaa_daily_summaries_USW00014740_2016_2025.json`,
  - `Heat Transfer Study/data/climate/noaa_heatwave_spell_anomaly_bounds_USW00014740_2016_2025.`
  - `Heat Transfer Study/tools/derive_noaa_heatwave_spell_bounds.py`.

4 Repository Scope Relevant To Section 18

Main execution code

- `Heat Transfer Study/heat_transfer_study.py`
- Section 18 core functions are concentrated at lines 5894–6740 (current repository state).

Main configuration inputs

- `Heat Transfer Study/study_config.json`
- Key blocks: `year_round_analysis`, `year_round_analysis.climate_data`, `cost_analysis`.

Primary generated outputs

- `Heat Transfer Study/outputs/year_round/study_summary.txt`
- `Heat Transfer Study/outputs/year_round/year_round_energy_cost_analysis.png`

Reference documentation

- `docs/execution_docs/physics/physics_and_math_reference.tex`
- Section 18 starts at line 1876 and includes Eqs. 229–251.

5 Execution Chain (What Happens In Code)

1. **Load and normalize annual config**: `year_round_analysis_config(...)` (L5894–).
2. **Build representative daily climate**: `build_representative_climate_year(...)` (L6158–), using:
   - `largest_remainder_allocation(...)` (L6009–),
   - `sample_truncated_normal(...)` (spell-level anomaly sampling),
   - `gaussian_weighted_day_selection(...)` (L6042–),
   - `heatwave_day_mask(...)` (L6082–),
   - `circular_gaussian_kernel_regression(...)` (L6141–).
3. **Run daily thermal and duty solve**: `simulate_year_round_day_task(...)` (L6389–).
4. **Aggregate annual and monthly outputs**: `build_year_round_payload(...)` (L6498–).
5. **Write report output text**: `write_year_round_summary(...)` (L6687–).

6 Variable Connection Map For Eqs. 229–251

Math Symbol	Equation Role	JSON Source	Code Variable / Expression	Output Field
$\bar{T}_m$	monthly mean ambient reference value	`climate_data.monthlyMeanC`	`monthly_mean`, `mean_c` (L6120, L6192)	`monthlyMean_C`
σ_m	monthly ambient spread reference value	`climate_data.monthlyStdC`	`monthly_std`, `std_c` (L6121, L6193)	`monthlyStd_C`, `ambientSigma_C`
h	Gaussian bandwidth (days)	`climate_data.gaussianKernelBandwidthDays`	`bandwidth_days` (L6169)	`gaussianBandwidth_days`
$\hat{y}(d)$	smooth daily baseline from reference values (Eq. 229)	same as above	`daily_mean_regressed` (L6171)	`ambientBase_C`
$\delta_i(d)$	circular day distance (Eq. 230)	period implied by days/month	`deltas = min(d, P-d)` (L6109–L6110)	not directly written
F_m	expected freeze days/month	`freeze.averageDaysPerMonth`	`freeze_expected` (L6124)	`freezeDaysExpected`
$n_{f,m}$	allocated integer freeze days (Eq. 231)	derived from F_m	`freeze_alloc` (L6157)	`freezeDaysAllocated`, annual freeze total
$\mu_{f,m}$	mean freeze day-of-month	`freeze.meanDayOfMonth`	`freeze_mean_day` (L6126)	`freezeMeanDayOfMonth`
$\Delta T_{f,m}$	freeze anomaly below monthly mean	`freeze.meanAnomalyBelowMonthlyMeanC`	`freeze_anomaly` (L6125)	`freezeAnomalyBelowMean_C`
$T_\infty(d)$ (freeze-adjusted)	freeze override (Eq. 234)	derived from baseline + anomaly	`ambient_profile[freeze_mask] = min(...)` (L6207–L6211)	`ambient_C`, `freezeDay`
S_m	expected heat-wave spell starts/month	`heat_wave.averageSpellStartsPerMonth`	`heat_expected_starts` (L6127)	`heatWaveSpellStartsExpec`
H_m	expected heat-wave days/month	`heat_wave.averageDaysPerMonth`	`heat_expected_days` (L6128)	`heatWaveDaysExpected`
$n_{s,m}, n_{h,m}$	allocated starts and days (Eq. 235)	derived from S_m, H_m	`heat_start_alloc`, `heat_day_alloc` (L6158–L6159)	`heatWaveSpellStartsAlloc` `heatWaveDaysAllocated`
$\mu_{h,m}$	mean heat-wave start day	`heat_wave.meanStartDayOfMonth`	`heat_mean_start` (L6130)	`heatWaveMeanStartDayOfMo`
$\Delta T_{h,m}$	monthly mean spell anomaly above monthly mean	`heat_wave.meanAnomalyAboveMonthlyMeanC` (NOAA-derived spell means)	`heat_anomaly_mean`	`heatWaveAnomalyAboveMean`
$\Delta T_{h,m}^{\text{low}}, \Delta T_{h,m}^{\text{high}}$	monthly spell-anomaly bounds	`heat_wave.minSpellAnomalyAboveMonthlyMeanC`, `maxSpellAnomalyAboveMonthlyMeanC` (loaded from derived NOAA bounds JSON)	`heat_anomaly_low`, `heat_anomaly_high`	`heatWaveAnomalyMin_C`, `heatWaveAnomalyMax_C`
$\sigma_{\Delta T,h,m}$	spell-anomaly sampling spread	`heat_wave.spellAnomalyStdC` (NOAA-derived spell spread; fallback from bounds if absent)	`heat_heat_std`	`heatWaveAnomalyStd_C`

Math Symbol	Equation Role	JSON Source	Code Variable / Expression	Output Field
$\Delta T_{h,m,j}$	sampled anomaly for spell j (bounded normal draw)	derived from monthly mean/bounds/std + seed	`sample_truncated_normal`	`heatWaveSpellAnomalies_C` day field `heatWaveSpellAnomalyAbov`
$T_\infty(d)$ (heat-adjusted)	heat-wave override (Eq. 238) with spell-specific anomaly	derived from baseline + $\Delta T_{h,m,j(d)}$	spell-loop max clamp on each selected spell window	`ambient_C`, `heatWaveDay`, `heatWaveSpellIndex`
$T_{h,\text{set}}$, $T_{c,\text{set}}$	heating/cooling setpoints	`year_round_analysis.heating_setpoint_C`, `coolingSetpoint_C` (or resolved defaults)	`heating_setpoint_C`, `cooling_setpoint_C` (L6388–L6389)	printed as resolved setpoints in summary
$Q_{\text{req},h}(d)$	required heating load (Eq. 239)	derived from bin model and ambient	`required_heat_W` (L6287)	`requiredHeat_W`
$Q_{\text{req},c}(d)$	required cooling load (Eq. 240)	derived from bin model and ambient	`required_cooling_W` (L6288)	`requiredCooling_W`
$Q_{h,\text{cap}}(d)$	heating capacity at day ambient (Eq. 241)	reference point + ambient re-solve	`nominal_capacity_W` from heating state (L6298–L6300)	`nominalCapacity_W`
$P_{h,\text{elec}}(d)$	heating electric power (Eq. 241)	same	`nominal_power_W` from `totalPower_W` (L6300)	`nominalPower_W`
$Q_{c,\text{cap}}(d)$	cooling capacity at day ambient (Eq. 242)	cooling point + ambient re-solve	`nominal_capacity_W = max(-QtoBed, 0)` (L6312)	`nominalCapacity_W`
$P_{c,\text{elec}}(d)$	cooling electric power (Eq. 242)	cooling solve outputs	`spotCoolerPower_W + assistBlowerPower_W` (L6313–L6314)	`nominalPower_W`
$D_h^*(d)$, $D_c^*(d)$	required duty fraction (Eq. 243)	derived ratio load/capacity	`duty_required` (L6320)	`dutyRequired`
$D_h(d)$, $D_c(d)$	clipped applied duty (Eqs. 244–245)	bounded physically to $[0, 1]$	`duty_applied` (L6321)	`dutyApplied`
$Q_{\text{bed}}(d)$	applied bed load (Eq. 246)	mode + duty + signed capacity	`q_bed_applied_W` (L6324–L6336)	`appliedQtoBed_W`
$T_b(d), T_t(d)$	bottom and top node temperatures (Eq. 247)	two-node model + applied bed load	`tb_C, tt_C` (L6326, L6331, L6336)	`bottom_C`, `top_C`
$E_h(d), E_c(d)$	daily mode-specific kWh (Eq. 248)	power, duty, 24 h conversion	`heating_kWh`, `cooling_kWh` (L6327, L6333)	`heating_kWh`, `cooling_kWh`
E_{day}	total daily kWh (Eq. 249)	sum of mode energies	`total_kWh` (L6340)	`total_kWh`
c_{elec}	electricity price (Eq. 250)	`cost_analysis.electricity_price_usd_per_kWh`	`electricity_price_usd_per_kWh` (L6390)	printed in annual summary
$C(d)$	daily cost (Eq. 250)	$c_{\text{elec}} E_{\text{day}}$	`cost_usd` (L6341)	`cost_usd`, cumulative cost
$m_{\text{water}}(d)$	duty-scaled daily water replacement (Eq. 251)	nominal water loss rate from day mode	`water_kg = water_rate_kg_day * duty_applied` (L6342)	`water_kg`

7 Sequential Variable Feed (Section 18 Format)

This section mirrors the Section 18 structure, but emphasizes the feed-forward dependency of variables.

18.1 Climate reference values and regression baseline (Eqs. 229–230)

Inputs:

- month lengths: L_m from `daysPerMonth`,
- month-midpoint reference days: d_i (equivalently d_m),
- monthly means: $\bar{T}_m$,
- monthly standard deviations: σ_m,
- kernel bandwidth: h.

Sequential feed:

$$\{\bar{T}_m,\ d_i,\ h\} \to \delta_i(d),\ w_i(d) \to \hat{T}_{\infty,\text{base}}(d)$$

$$\{\sigma_m,\ d_i,\ h\} \to \delta_i(d),\ w_i(d) \to \hat{\sigma}(d)$$

Outputs passed forward:

- baseline daily ambient: $T_{\infty,\text{base}}(d)$,
- daily sigma context: $\sigma(d)$.

18.2 Freeze-day allocation and anomaly override (Eqs. 231–234)

Inputs:

- expected freeze days F_m,
- mean freeze day $\mu_{f,m}$,
- freeze anomaly center $M_{f,m} \equiv \Delta T_{f,m}$,
- freeze anomaly bounds $\Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}$,
- freeze anomaly spread $\sigma_{\Delta T,f,m}$,
- freeze anomaly model key κ_f,
- freeze anomaly RNG seed s_f,
- $T_{\infty,\text{base}}(d)$ from 18.1.

Sequential feed:

$$F_m \to n_{f,m} = \text{LR}(F_m)$$

$$\{n_{f,m}, \mu_{f,m}\} \to w_{f,m}(i),\ \tilde{w}_{f,m}(i) \to \mathcal{S}_{f,m}$$

Here $s_{f,m}$ is the month-m freeze-day spread parameter and $r_{f,m}$ is the repulsion length scale.

$$a_{f,m} = \Delta T^{\text{low}}_{f,m}, \quad b_{f,m} = \Delta T^{\text{high}}_{f,m}, \quad \mu^{\star}_{f,m} = \text{clip}(M_{f,m}, a_{f,m}, b_{f,m}).$$

$$c_{f,m} = \mathrm{clip}\Big(3\mu^{\star}_{f,m} - a_{f,m} - b_{f,m},\ a_{f,m},\ b_{f,m}\Big).$$

$$t_{f,m}(x) = \begin{cases} \dfrac{2(x - a_{f,m})}{(b_{f,m} - a_{f,m})(c_{f,m} - a_{f,m})}, & a_{f,m} \le x \le c_{f,m}, \\ \dfrac{2(b_{f,m} - x)}{(b_{f,m} - a_{f,m})(b_{f,m} - c_{f,m})}, & c_{f,m} \le x \le b_{f,m}, \\ 0, & \text{otherwise.} \end{cases}$$

$$\mathcal{D}_{\mathrm{det}} = \{\texttt{off}, \texttt{none}, \texttt{disabled}, \texttt{deterministic}, \texttt{deterministic_mean}\},$$

$$\mathcal{D}_{\triangle} = \{\texttt{triangular}, \texttt{triangular_mean_matched}, \texttt{triangular_between_monthly_min_max}, \texttt{triangular_mean_mat}$$

$$D_{\triangle} = \mathcal{D}_{\triangle} = \{\texttt{triangular}, \texttt{triangular_mean_matched}, \texttt{triangular_between_monthly_min_max}, \texttt{triangular_me}$$

$$\Delta T_{f,m,q} = \begin{cases} \mu^{\star}_{f,m}, & & \kappa_f \in \mathcal{D}_{\mathrm{det}}, \\ X_{f,m,q}, & X_{f,m,q} = \mathrm{Triangular}(a_{f,m}, c_{f,m}, b_{f,m}), & \kappa_f \in \mathcal{D}_{\triangle} \text{ and } b_{f,m} > a_{f,m}, \\ \mu^{\star}_{f,m}, & & \text{otherwise.} \end{cases}$$

$X_{f,m,q} = \mathrm{Triangular}(a_{f,m}, c_{f,m}, b_{f,m})$ for the default freeze model, then clip to $[a_{f,m}, b_{f,m}]$.

$$\Delta T_{f,m,q} \leftarrow \mathrm{clip}(\Delta T_{f,m,q}, a_{f,m}, b_{f,m}).$$

Sort selected freeze days as $s_{f,m,1} < \cdots < s_{f,m,n_{f,m}}$,

$$A_{f,m}(i) = \begin{cases} 0, & i \notin \mathcal{S}_{f,m}, \\ \Delta T_{f,m,q}, & i = s_{f,m,q} \text{ for some } q \in \{1, \ldots, n_{f,m}\}, \end{cases} \qquad i = 1, \ldots, L_m.$$

$$i_m(d) = \text{day-of-month index for day-of-year } d \text{ in month } m.$$

$$\{T_{\infty,\mathrm{base}}(d), \bar{T}_m, \Delta T_{f,m,q}, \mathcal{S}_{f,m}\} \to T_{\infty,\mathrm{freeze}}(d) = \min\Big(T_{\infty,\mathrm{base}}(d), \bar{T}_m - A_{f,m}(i_m(d))\Big)$$

Output passed forward:

- freeze-adjusted ambient profile.

18.3 Heat-wave allocation and anomaly override (Eqs. 235–238)

Inputs:

- expected heat-wave starts S_m,
- expected heat-wave days H_m,
- mean heat-wave start day $\mu_{h,m}$,
- heat-wave mean anomaly $\Delta T_{h,m}$,
- heat-wave anomaly bounds $\Delta T^{\mathrm{low}}_{h,m}, \Delta T^{\mathrm{high}}_{h,m}$,
- heat-wave anomaly spread $\sigma_{\Delta T,h,m}$,
- freeze-adjusted ambient from 18.2.

Stepwise derivation:

$$\text{Step 1 (integer allocation):} \quad \{S_m, H_m\} \to \{n_{s,m}, n_{h,m}\} = \{\mathrm{LR}(S_m), \mathrm{LR}(H_m)\}.$$

$$\text{Step 2 (spell lengths):} \quad \ell_{h,m,j} = 3,\ j = 1, \ldots, n_{s,m}, \qquad e_m = n_{h,m} - 3n_{s,m},$$

then distribute e_m extra days across spells to obtain $\boldsymbol{\ell}_{h,m} = (\ell_{h,m,1}, \ldots, \ell_{h,m,n_{s,m}})$ with $\sum_j \ell_{h,m,j} = n_{h,m}$.

$$\text{Step 3 (start-day base score):} \quad w_{h,m}(i) = \exp\left[-\frac{(i-\mu_{h,m})^2}{2s_{h,m}^2}\right], \qquad i = 1, \ldots, L_m.$$

$$\text{Step 4 (recursive selected-start set):} \quad \mathcal{S}_{h,m}^{(0)} = \varnothing,$$

$$\mathcal{I}_{h,m}^{(k)} = \{1, \ldots, L_m\} \setminus \mathcal{S}_{h,m}^{(k-1)},$$

$$\mathrm{score}_{h,m}^{(k)}(i) = w_{h,m}(i) \prod_{j \in \mathcal{S}_{h,m}^{(k-1)}} \left[1 - 0.65 \exp\left(-\frac{(i-j)^2}{2r_{h,m}^2}\right)\right], \qquad i \in \mathcal{I}_{h,m}^{(k)},$$

$$i_k \in \mathcal{I}_{h,m}^{(k)}, \qquad \mathrm{score}_{h,m}^{(k)}(i_k) = \max_{i \in \mathcal{I}_{h,m}^{(k)}} \mathrm{score}_{h,m}^{(k)}(i),$$

$$\mathcal{S}_{h,m}^{(k)} = \mathcal{S}_{h,m}^{(k-1)} \cup \{i_k\}, \qquad k = 1, \ldots, n_{s,m}, \qquad \mathcal{S}_{h,m} = \mathcal{S}_{h,m}^{(n_{s,m})}.$$

Here $s_{h,m}$ is month-m start spread and $r_{h,m}$ is the repulsion length scale. At iteration k, "already selected" means exactly $\mathcal{S}_{h,m}^{(k-1)}$.

Step 5 (feasible contiguous spells): project raw starts to feasible $s_{h,m,j}$ so spells do not overlap and remain in m

$$\mathcal{B}_{h,m,j} = \{\, d :\ s_{h,m,j} \le d \le s_{h,m,j} + \ell_{h,m,j} - 1 \,\}$$

$$\text{and } j(d) = j \text{ when } d \in \mathcal{B}_{h,m,j}.$$

Step 6 (spell anomaly sampling and ambient override): $\Delta T_{h,m,j} \in [a_{h,m}, b_{h,m}]$, $\Pr(u \le \Delta T_{h,m,j} \le v) = \int_u^v f_{h,m}$

$$\text{for } a_{h,m} \le u \le v \le b_{h,m}.$$

$$a_{h,m} = \Delta T_{h,m}^{\mathrm{low}}, \quad b_{h,m} = \Delta T_{h,m}^{\mathrm{high}}, \quad \mu_{h,m}^{\star} = \mathrm{clip}(\Delta T_{h,m}, a_{h,m}, b_{h,m}), \quad \sigma_{h,m}^{\star} = \max(\sigma_{\Delta T,h,m}, \varepsilon),$$

$$g_{\mathrm{std}}(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right), \qquad G_{\mathrm{std}}(z) = \int_{-\infty}^{z} g_{\mathrm{std}}(t)\, dt.$$

$$f_{h,m}(x) = \frac{g_{\mathrm{std}}\left(\frac{x-\mu_{h,m}^{\star}}{\sigma_{h,m}^{\star}}\right)}{\sigma_{h,m}^{\star}\left[G_{\mathrm{std}}\left(\frac{b_{h,m}-\mu_{h,m}^{\star}}{\sigma_{h,m}^{\star}}\right) - G_{\mathrm{std}}\left(\frac{a_{h,m}-\mu_{h,m}^{\star}}{\sigma_{h,m}^{\star}}\right)\right]}, \quad a_{h,m} \le x \le b_{h,m},$$

$$\text{and } f_{h,m}(x) = 0 \text{ outside } [a_{h,m}, b_{h,m}].$$

$$\underbrace{f_{h,m}(x)}_{\text{distribution}} \Rightarrow \underbrace{\Delta T_{h,m,j}}_{\text{sample for spell } j} \Rightarrow \underbrace{T_{h,m,j}^{\mathrm{th}} = \bar{T}_m + \Delta T_{h,m,j}}_{\text{spell threshold}} \Rightarrow \underbrace{T_\infty(d)}_{\text{daily ambient passed to 18.4}}.$$

$$\{T_{\infty,\mathrm{freeze}}(d), \bar{T}_m, \Delta T_{h,m,j(d)}\} \to T_\infty(d) = \max\Big(T_{\infty,\mathrm{freeze}}(d), \bar{T}_m + \Delta T_{h,m,j(d)}\Big)$$

Output passed forward:

- final daily ambient $T_\infty(d)$ used by all day-level thermal solves.

18.4 Daily heating and cooling loads (Eqs. 239–240)

Inputs:

- $T_\infty(d)$ from 18.3,
- heating and cooling setpoints $T_{h,\text{set}}, T_{c,\text{set}}$,
- day-specific bin thermal model terms (effective UA_{bin}).

Sequential feed:

$$\{T_\infty(d), T_{h,\text{set}}, UA_{\text{bin}}\} \to Q_{\text{req},h}(d)$$
$$\{T_\infty(d), T_{c,\text{set}}, UA_{\text{bin}}\} \to Q_{\text{req},c}(d)$$

Outputs passed forward:

- required day heating/cooling loads.

18.5 Capacity re-solve and duty application (Eqs. 241–246)

Inputs:

- $Q_{\text{req},h}(d)$, $Q_{\text{req},c}(d)$ from 18.4,
- final ambient $T_\infty(d)$ from 18.3,
- selected heating and cooling reference points.

Sequential feed:

$$\{T_\infty(d), \text{heating reference point}\} \to \{Q_{h,\text{cap}}(d), P_{h,\text{elec}}(d)\}$$
$$\{T_\infty(d), \text{cooling reference point}\} \to \{Q_{c,\text{cap}}(d), P_{c,\text{elec}}(d)\}$$
$$\{Q_{\text{req},h}(d), Q_{h,\text{cap}}(d)\} \to D_h^*(d) \to D_h(d) = \text{clip}_{[0,1]}(D_h^*)$$
$$\{Q_{\text{req},c}(d), Q_{c,\text{cap}}(d)\} \to D_c^*(d) \to D_c(d) = \text{clip}_{[0,1]}(D_c^*)$$
$$\{D_h(d), D_c(d), Q_{h,\text{cap}}(d), Q_{c,\text{cap}}(d)\} \to Q_{\text{bed}}(d)$$

Outputs passed forward:

- applied bed thermal load $Q_{\text{bed}}(d)$,
- applied duty fraction (D_h or D_c),
- mode-specific electric power ($P_{h,\text{elec}}$ or $P_{c,\text{elec}}$).

18.6 Daily temperatures, electricity, cost, and water (Eqs. 247–251)

Inputs:

- $Q_{\text{bed}}(d)$, duty, and mode power from 18.5,
- $T_\infty(d)$ from 18.3,
- two-node conductance parameters and f_b,
- electricity unit price c_{elec},
- nominal daily water-loss rate $\dot{m}_{\text{water},24h}$.

Sequential feed:

$$\{Q_{\text{bed}}(d), T_\infty(d), UA_{b,\infty}, UA_{t,\infty}, UA_{bt}, f_b\} \to \{T_b(d), T_t(d)\}$$

$$\{P_{h,\text{elec}}(d), D_h(d)\} \to E_h(d), \qquad \{P_{c,\text{elec}}(d), D_c(d)\} \to E_c(d) \to E_{\text{day}}$$

$$\{E_{\text{day}}, c_{\text{elec}}\} \to C(d)$$

$$\{\dot{m}_{\text{water},24h}, D(d)\} \to m_{\text{water}}(d)$$

Final annual aggregation feed:

$$\sum_{d=1}^{365} \{E_h(d), E_c(d), C(d), m_{\text{water}}(d)\} \to \{\text{annual kWh, annual USD, annual water}\}$$

8 Clear Report Narrative You Can Follow

Use this 8-step order when drafting your report section:

1. **State the objective**: "Section 18 converts monthly climate statistics into a representative daily operating year and computes day-level thermal duty, electricity, cost, and water."
2. **Define climate reference values**: monthly means $\bar{T}_m$ and spreads σ_m from NOAA-derived arrays.
3. **Describe baseline reconstruction**: explain periodic Gaussian kernel regression and why circular distance removes Dec/Jan discontinuity.
4. **Describe event overlays**: explain freeze-day and heat-wave day integer allocation (largest remainder), day placement (Gaussian-centered with repulsion), and spell-level bounded-normal anomaly sampling for heat-wave intensity.
5. **Move to daily thermal demand**: compute required heating/cooling from day ambient and setpoints.
6. **Explain ambient-specific capacity re-solving**: heating and cooling reference points are re-evaluated at each day ambient before duty is computed.
7. **Explain bounded operation**: duty is clipped to $[0, 1]$, applied bed load is signed by mode, and unmet load is tracked separately when needed.
8. **Close with economics and resources**: convert day power to kWh, then to cost (USD/day), and duty-scaled water replacement, then aggregate by month and year.

9 Draft Paragraph Template (Copy And Adapt)

Section 18 implements an equation-driven annual operating layer that maps monthly climate reference values to daily ambient conditions, then computes daily heating or cooling requirements and operating costs. First, monthly mean and standard-deviation reference values are converted into a smooth 365-day baseline using periodic Gaussian kernel regression (Eqs. 229–230). Next, freeze days and heat-wave spells are allocated by largest-remainder rounding and positioned with Gaussian timing weights plus repulsion to avoid unrealistic clustering; heat-wave spell magnitudes are then drawn from seeded bounded-normal monthly distributions (Eqs. 231–238). For each day, required heating/cooling loads are evaluated from the day ambient and setpoints (Eqs. 239–240). The model then re-solves heating and cooling reference operating points at that specific ambient, computes required duty, clips applied duty to physical bounds, and applies signed bed load by mode (Eqs. 241–246). Bottom and top bed-node temperatures are solved from the same two-node steady model used elsewhere (Eq. 247), and daily electricity, cost, and water replacement are computed and aggregated to monthly and annual totals (Eqs. 248–251).

10 Quality Checks To Mention In The Report

- Duty clipping enforces physical feasibility of operation (`dutyApplied` in $[0, 1]$).
- Unmet load is explicitly quantified (`unmetLoad_W`, `annualUnmet_kWh_equivalent`).
- Heating and cooling power are ambient-specific daily re-solves, not fixed constants.
- Freeze/heat-wave event totals are auditable at monthly and annual levels.
- Heat-wave spell anomalies are bounded by configured monthly low/high arrays and reproducible via fixed RNG seed.

11 Current Run Snapshot (Example Numbers)

From `Heat Transfer Study/outputs/year_round/study_summary.txt` in this repository state:

- Allocated annual freeze days: 120
- Allocated annual heat-wave starts: 4
- Allocated annual heat-wave days: 15
- Annual electricity: 660.9 kWh
- Annual operating cost: 190.01 USD
- Annual water replacement: 3.4 kg

Use these only as run-specific examples; your final report values should come from the run you cite.